Melting Temperature Estimation: Wallace Rule vs Nearest-Neighbor Model

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1 Introduction

The melting temperature (T_m) can be defined as the temperature at which 50% of a DNA sequence is double-stranded (dsDNA), while the other 50% is single-stranded (ssDNA).

In molecular assays, measuring the right T_m is crucial. There have been different methods to determine T_m. Simpler methods consider only the content of bases in a sequence, such as the Wallace rule. However, there are many variables that affect T_m, which are considered in the Nearest-Neighbor model (NN) and its corrections, currently the standard for determining T_m.

This small analysis, through simulations, focused on exploring the differences in the T_m calculation between the Wallace rule and NN model with salt correction.

2 Methods

The simulations were developed with the R programming language (in the RStudio environment). First, three groups of primers were simulated: small (12-17 bp), medium (18-24 bp), and large (25-30 bp) (Table 1). For each group, 1000 primers were generated (Table 1).

Group	Length (bp)	Simulated primers
Small	12–17	1000
Medium	18–24	1000
Large	25–30	1000

Table 1: Primer groups used for simulation

The first method to calculate the T_m was the Wallace rule:

$$T_m = 4(G + C) + 2(A + T)$$

This rule considers only the absolute frequency of the bases in a sequence.

The second method to determine T_m was NN with salt correction:

$$T_m = \frac{\Delta H^\circ}{\Delta S^\circ + R \ln\left(\frac{C_t}{4}\right)} - 273.15 + 16.6 \log_{10}[\mathrm{Na}^+]$$

This formula relies on thermodynamic parameters derived from the nearest-neighbor model, which considers the dinucleotide composition of the DNA sequence (SantaLucia et al., 1996). The ΔH° and ΔS° values were obtained from a unified parameter set (SantaLucia, 1998), and the effect of salt concentration on T_m was incorporated using an empirical correction (Schildkraut & Lifson, 1965). Primer and salt concentrations were set to 0.25 μM and 50 mM, respectively.

After determining T_m for the three primer groups with both methods, the difference in T_m (ΔT_m) and $|\Delta T_m|$ were calculated.

3 Results

The increase in $|\Delta T_m|$ is proportional to the increase in primer length (Figure 1). The large primer group had higher values of $|\Delta T_m|$, the lowest $|\Delta T_m|$ values were observed in the small primer group (Figure 1).

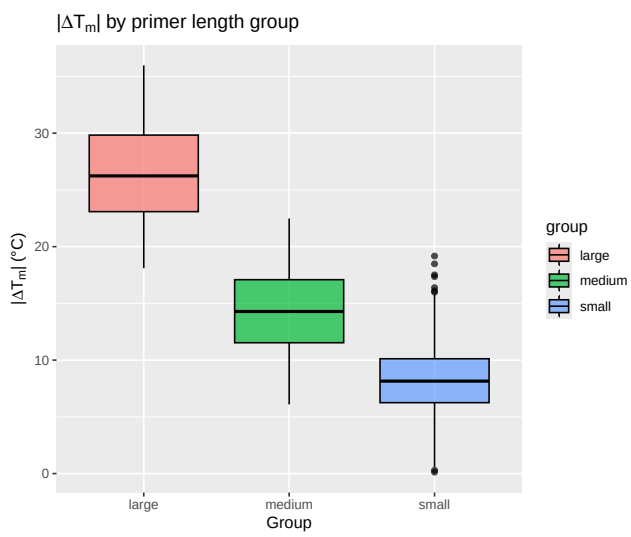

Figure 1: Absolute difference in T_m ($|\Delta T_m|$) by primer length group

In all of the primer groups, the T_m calculated by NN was generally lower than the Wallace rule output (Figure 2). However, values of both methods became closer when the T_m was around 50 °C (Figure 2).

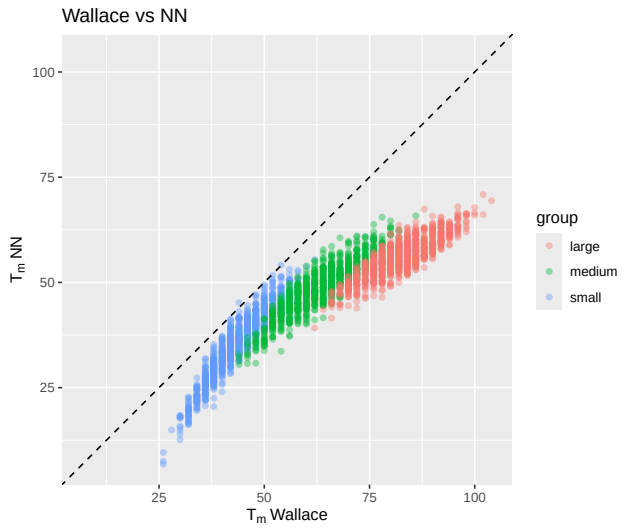

Figure 2: Wallace vs NN T_m values

In none of the primer groups was ΔT_m centered around 0 (Figure 3). Only in the small primer group were a few values observed close to 0 (Figure 3).

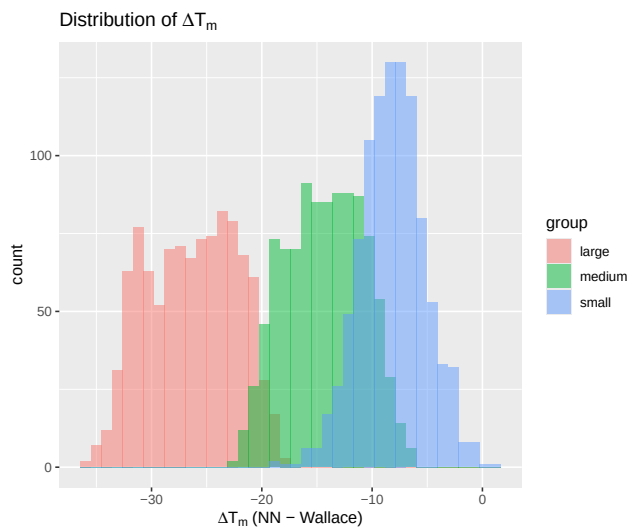

Figure 3: Distribution of ΔT_m

The large primer group showed a mean $|\Delta T_m|$ of 26.38 with a 95% confidence interval (CI) of (26.13, 26.63) (Table 2). Similarly, the medium and small primer groups had mean values of 14.31 (CI: 14.09, 14.53) and 8.14 (CI: 7.95, 8.32), respectively (Table 2). None of the confidence intervals include 0, indicating that the difference between the two methods is consistently different from zero across all primer length groups.

Group	$\|\Delta T_m\|$ Mean	$\|\Delta T_m\|$ 95% CI
Large	26.38	(26.13, 26.63)
Medium	14.31	(14.09, 14.53)
Small	8.14	(7.95, 8.32)

Table 2: Mean and 95% confidence interval of $|\Delta T_m|$ for each primer group

4 Limitation

The NN model in this analysis considered a simple salt correction for [Na^+]. However, more complex corrections of the NN model include [dNTPs], [Mg^{2+}] or secondary structure formation. Particularly, [Mg^{2+}] tends to elevate T_m.

Also, primer groups were determined only by length, but generating groups by variables like %GC might help to improve the analysis.

5 Conclusion

The T_m measured by both methods seem to differ significantly. While the Wallace rule seems to overestimate T_m, the NN model with basic salt correction might be underestimating T_m because other factors were not considered. The Wallace rule has a role in learning environments thanks to its simplicity, while the NN model can achieve more precise measurements.

6 References

- SantaLucia, J., Jr., Allawi, H. T., & Seneviratne, P. A. (1996). Improved nearest-neighbor parameters for predicting DNA duplex stability. *Biochemistry*, 35(11), 3555–3562. `https://doi.org/10.1021/bi951907q`
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- Schildkraut, C., & Lifson, S. (1965). Dependence of the melting temperature of DNA on salt concentration. *Biopolymers*, 3(2), 195–208. `https://doi.org/10.1002/bip.360030207`